

GIS Data Governance Framework

A practical operating model for reliable enterprise geospatial data

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[GIS Leadership Portfolio](#)

What this sample demonstrates

- Governance policy translated into decision rights and repeatable operating controls
- A geospatial data lifecycle with measurable acceptance and publication gates
- Metadata, quality, access, change, exception, and retirement controls
- A phased implementation approach designed to scale with program maturity

Document status

This is a generalized and sanitized portfolio sample. Organization names, personnel, locations, systems, and operational details have been removed. It is not an adopted policy and should be tailored before operational use.

A governance system people can actually use

A GIS program becomes dependable when ownership, authority, quality expectations, access, and lifecycle responsibilities are explicit. This framework connects those decisions to evidence records and measurable performance without creating unnecessary bureaucracy.

Core principles

Principle	Working rule
Mission aligned	Prioritize controls that protect a defined business decision, service, or obligation.
Authoritative by design	Assign one accountable owner and document the approved system of record.
Fit for use	Set quality thresholds according to intended use, risk, and lifecycle tier.
Least privilege	Give users only the access needed for their responsibilities.
Traceable change	Record material changes, findings, exceptions, approvals, and disposition decisions.
Reuse before recreation	Evaluate existing governed content before creating another copy.

Governance tiers

Tier	Use	Control level
Critical	Authoritative operational or regulatory data	Formal validation, controlled change, scheduled review
Managed	Shared program data and reusable services	Named owner, metadata, quality review, role-based access
Working	Project and analytical data	Labeled limitations and a retention or promotion decision
Public	Approved open data and public applications	Publication, sensitivity, privacy, and licensing review

Governance operating model

The model separates business authority, technical administration, custodianship, and oversight so that decisions do not default to whoever happens to manage the GIS platform.

Role	Accountability	Decisions
Executive sponsor	Direction, resources, escalation	Policy, priorities, material exceptions
GIS program lead	Governance process and performance	Standards, reviews, coordination, escalation
Data owner	Purpose, authority, acceptable use	Quality thresholds, access, publication, retirement
Data custodian	Maintenance, metadata, quality	Updates, remediation, lifecycle records
Platform administrator	Security, availability, recovery	Roles, permissions, logging, technical controls
Producer	Conformance of submitted data	Corrections and source documentation
Consumer	Appropriate use and issue reporting	Use within limits and report suspected defects
Governance council	Cross-functional review	Ownership, access, standards, investment conflicts

Key decision rights

Decision	Accountable role	Required consultation
Declare content authoritative	Data owner	GIS lead, custodian, affected business units
Approve external or public release	Data owner	Security or privacy reviewer, GIS lead
Change a shared schema or service	Data owner	Custodian, administrator, affected consumers
Grant elevated access	Platform administrator	Data owner and GIS lead
Accept a time-bound exception	Executive sponsor or delegate	Owner, GIS lead, risk stakeholder

From intake through retirement

Each lifecycle stage has a control objective, a decision gate, and an evidence record. Controls become stronger as the data tier and business risk increase.

01 Identify	02 Acquire	03 Validate	04 Publish	05 Maintain	06 Retire
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Stage	Control objective	Evidence
Identify	Confirm the need, intended decision, owner, and whether suitable data already exists.	Intake record and tier
Acquire	Document source, method, license, specification, and delivery expectations.	Source record or specification
Validate	Test spatial, thematic, structural, metadata, and sensitivity requirements.	Checklist and findings log
Publish	Confirm acceptance, access, configuration, metadata, and user limitations.	Approval and publication record
Maintain	Monitor currency, dependencies, use, incidents, and reported defects.	Maintenance log and measures
Retire	Remove obsolete content while preserving required history and resolving dependencies.	Disposition and archive record

Acceptance outcomes

Outcome	Definition
Pass	Required controls meet the approved acceptance criteria.
Conditional pass	Minor findings are assigned, dated, and accepted by the owner.
Reject	Material quality, security, licensing, or usability issues prevent acceptance.
Exception	A designated authority accepts a time-bound deviation with compensating controls.

Minimum metadata and validation controls

Metadata and QA/QC are treated as operating controls, not optional documentation. The required depth should match the data tier and intended use.

Metadata area	Minimum information
Identity	Title, summary, theme, geographic extent, identifier, keywords
Authority	Owner, custodian, authoritative status, source, system of record
Lineage	Source data, collection method, transformations, dependencies
Quality	Accuracy, completeness, validation date, findings, fitness for use
Time	Creation, last update, update cycle, next review, temporal coverage
Access	Sensitivity, approved audience, constraints, license, sharing limits
Technical	Geometry, coordinate system, schema version, service or repository
Lifecycle	Tier, status, retention, archive location, retirement date

Core validation dimensions

Dimension	Representative checks
Spatial	Coordinate system, geometry validity, alignment, topology, extent, scale
Thematic	Required fields, domains, units, ranges, nulls, outliers, duplicates
Structural	Schema, field types, naming, relationships, keys, service compatibility
Completeness	Expected features, attributes, time periods, coverage, deliverables
Currency	Update date and refresh cycle support the intended use
Documentation	Lineage, accuracy, limitations, contact, and constraints are complete
Security	Sensitivity, privacy, proprietary limits, access, and audience are verified

Access publication and material change

Access classes

Class	Typical permission	Approval basis
Administrator	Platform and security administration	Assigned duty and elevated-access review
Custodian	Maintain assigned governed content	Named custodianship
Contributor	Submit or edit scoped content	Business need, training, bounded access
Internal viewer	View and query approved internal content	Need to know and acceptable use
External partner	Time-bound access to approved content	Agreement, sponsor, scope, expiration
Public	Approved public products only	Publication and sensitivity review

Publication gate

- Confirm ownership, approved audience, and authoritative status.
- Check privacy, sensitivity, proprietary, licensing, and aggregation risks.
- Verify metadata, limitations, update frequency, contact, and accessibility.
- Test service behavior, permissions, pop-ups, downloads, and dependent applications.
- Record approval, publication date, reviewer, dependencies, and next review date.

Material changes and exceptions

Changes to a shared schema, coordinate system, service endpoint, field meaning, domain, update method, access class, or authoritative status require impact review. The record identifies the rationale, affected consumers, test results, rollback plan, approvals, deployment date, and communications.

Exceptions must be specific, time-bound, risk-based, and owned. Each record states the unmet requirement, business justification, risk, compensating controls, approver, expiration, and corrective action.

Implementation roadmap and scorecard

Phase	Focus	Tangible output
1 Establish	Sponsor, scope, principles, roles, priorities	Charter, inventory, ownership, baseline
2 Control	Intake, metadata, validation, access, change	Templates, checklists, role matrix
3 Pilot	Apply controls to high-value datasets	Validated data, findings, lessons learned
4 Scale	Expand coverage and automate repeatable checks	Training, dashboards, scheduled reviews
5 Improve	Use measures and feedback to refine controls	Policy updates and targeted investment

Governance scorecard

Measure	Definition	Direction
Ownership coverage	Governed datasets with a named owner and custodian	Increase
Metadata completeness	Datasets meeting the minimum profile	Increase
Validation pass rate	Submissions accepted without material rework	Increase
Finding closure time	Median days to close quality findings	Decrease
Freshness compliance	Datasets updated within the approved cycle	Increase
Duplicate reduction	Redundant datasets retired or consolidated	Improve
Access review completion	Scheduled access reviews completed on time	Increase
Service reliability	Availability and successful refresh rate	Increase

Reusable evidence records

Dataset intake record | validation checklist | quality findings log | publication approval | access review | change record | exception record | retirement record

This sample shows how governance moves from policy language into operating decisions, evidence, and measurable performance. The editable version can be adapted for a specific organization after its legal, security, records, and platform requirements are confirmed.